

学生作答详细校对

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当前为已进入后续阶段的历史回看；学生身份与识别作答保持只读。

身份状态 待确认 文件名匹配	作答覆盖 100% 7/7	待复核题次 0 道题	来源文件 1 份文件
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← 寮狄竺

[Unknown Student] AbstractAlgebraHw1.pdf

下一位学生 →

✎ 修正身份

Q SmarTAI 智能搜索：题号、题型或“积分题”等

搜索只筛选题目：当前学生保持不变。输入中文时会在选词完成后再用筛选。
🗎 输入框聚焦时方向键只编辑文本；退出输入框后，左右切学生、上下切题目。

- 题目导航
- 共 7 题 · 点击定位
- 第 1.1.5 题
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- 第 1.1.31 题
- 第 1.2.3 题
- 第 1.2.16 ...

第 1.1.5 题 证明题 已识别

题目与学生作答在同一卡片内连续校对；题目只读。

↑ 上一题 下一题 ↓

题目上下文

Prove for all $n > 1$ that $\mathbb{Z}/n\mathbb{Z}$ is not a group under multiplication of residue classes.

学生作答识别文本

浏览状态渲染 LaTeX；需要修正时再编辑原始文本。

hw/AbstractAlgebraHw1.pdf

Suppose that $\mathbb{Z}/n\mathbb{Z}$ is, let e be the identity, denote e as $e' + n\mathbb{Z}$ ($e' \in \mathbb{Z}$).
thus $(1 + n\mathbb{Z}) \times (e' + n\mathbb{Z}) = 1 + n\mathbb{Z}$ implies $e' \equiv 1 \pmod n$
let $b + n\mathbb{Z}$ be the inverse of $a + n\mathbb{Z}$, thus $b \equiv e' \equiv 1 \pmod n$, Contradict.
thus $\mathbb{Z}/n\mathbb{Z}$ isn't a multi. group.

第 1.1.7 题 证明题 已识别

题目与学生作答在同一卡片内连续校对；题目只读。

↑ 上一题 下一题 ↓

题目上下文

Let $G = \{x \in \mathbb{R} \mid 0 \leq x < 1\}$ and for $x, y \in G$ let $x * y$ be the fractional part of $x + y$ (i.e., $x * y = x + y - [x + y]$ where $[a]$ is the greatest integer less than or equal to a).
Prove that $*$ is a well defined binary operation on G and that G is an abelian group under $*$ (called the real numbers mod 1).

学生作答识别文本

浏览状态渲染 LaTeX；需要修正时再编辑原始文本。

hw/AbstractAlgebraHw1.pdf

Firstly. $x * y = \{x + y\} \in [0, 1) = G$, thus operation $*$ is conserved by G
 $(x * y) * z = x + y - [x + y] + z = [x + y - [x + y] + z] = x + y + z - [x + y + z]$
 $= x + y + z - [y + z] + [x + y + z - [y + z]] = x * (y * z)$.
thus $*$ is associative.
 $(x * y) = x + y - [x + y] = y * x$, thus $*$ is commutative.
Secondly. the identity is 0. because $x * 0 = x + 0 - [x] = x$ ($0 * x = x * 0 = x$, definitely)
for $\forall x$, its inverse is $\begin{cases} 1 - x, & x > 0 \\ 0, & x = 0 \end{cases}$, obviously.
To sum up, G is a well-defined Abelian group in terms of $*$.

第 1.1.20 题 证明题 已识别

题目与学生作答在同一卡片内连续校对；题目只读。

↑ 上一题 下一题 ↓

题目上下文

For x an element in G show that x and x^{-1} have the same order.

学生作答识别文本

浏览状态渲染 LaTeX；需要修正时再编辑原始文本。

hw/AbstractAlgebraHw1.pdf

Suppose that the order of x is n , that is, n is the smallest t s.t. $x^t = e$.
thus. $(x^{-1})^n = x^{-n}$ (due to $(x^n)^{-1} = (x^{-1})^n$)
 $(x^{-1})^n = (x^{-1})^n x^n = x^{-1} x^{-1} \dots x^{-1} x x x \dots x = e$
Also, suppose that $(x^{-1})^t = e$, by the proposition that $(x^{-1})^{-1} = x$
and the same reasoning as the previous one, we have $x^t = e$, thus $t \geq n$.
Sequentially, the order of x^{-1} is also n .

第 1.1.29 题 证明题 已识别

题目与学生作答在同一卡片内连续校对；题目只读。

↑ 上一题 下一题 ↓

题目上下文

Prove that $A \times B$ is an abelian group if and only if both A and B are abelian.

学生作答识别文本

浏览状态渲染 LaTeX；需要修正时再编辑原始文本。

hw/AbstractAlgebraHw1.pdf

If $A \times B$ is Abelian $\iff \forall (a_1, b_1), (a_2, b_2) \in A \times B, (a_1 a_2, b_1 b_2) = (a_1, b_1)(a_2, b_2)$.
 $\iff \forall a_1, a_2 \in A, b_1, b_2 \in B, a_1 a_2 = a_2 a_1$ and $b_1 b_2 = b_2 b_1$
 $\iff A, B$ are both Abelian.

第 1.1.31 题 证明题 已识别

题目与学生作答在同一卡片内连续校对；题目只读。

↑ 上一题 下一题 ↓

题目上下文

Prove that any finite group G of even order contains an element of order 2. [Let $t(G)$ be the set $\{g \in G \mid g = ft \text{ } g^{-1}\}$. Show that $t(G)$ has an even number of elements and every nonidentity element of $G - t(G)$ has order 2.]

学生作答识别文本

浏览状态渲染 LaTeX；需要修正时再编辑原始文本。

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Suppose that $T = \{x \in G \mid x \neq x^{-1}\}$. thus $\forall x \in T, x^{-1} \in T$
therefore the finite set T can be seperated into several pairs in form of
 $(a, a^{-1}) \Rightarrow |G - T|$ is even.
 $G - T$ has more than 1 element, thus there exists $a \neq e \in T$,
hence that $a = a^{-1}$, thus $a^2 = aa^{-1} = e \Rightarrow a$ has order 2.

第 1.2.3 题 证明题 已识别

题目与学生作答在同一卡片内连续校对；题目只读。

↑ 上一题 下一题 ↓

题目上下文

Use the generators and relations above to show that every element of D_{2n} which is not a power of r has order 2. Deduce that D_{2n} is generated by the two elements s and sr , both of which have order 2.

学生作答识别文本

浏览状态渲染 LaTeX；需要修正时再编辑原始文本。

hw/AbstractAlgebraHw1.pdf

$\forall x \in D_{2n}$ that is not a power of r , denote that $x = sr^\alpha$.
thus $x^2 = sr^\alpha sr^\alpha = s(sr^{-\alpha})r^\alpha = s^2 = 1$.
that is, the order of x is 2 because $x \neq 1$.

第 1.2.16 题 证明题 已识别

题目与学生作答在同一卡片内连续校对；题目只读。

↑ 上一题 下一题 ↓

题目上下文

Show that the group $\langle X, Y \mid X^2 = Y^2 = (XY)^2 = 1 \rangle$ is the dihedral group D_4 (where X, Y may be replaced by the letter r and y by s). [Show that the last relation is the same as: $X^2 Y = Y X^2$]

学生作答识别文本

浏览状态渲染 LaTeX；需要修正时再编辑原始文本。

hw/AbstractAlgebraHw1.pdf

Replace x_i by r, y_i by s we get $r^2 = s^2 = 1, r^2 s^2 = 1$
What's more, $rs = (rs)^{-1} \dots = sr \Rightarrow rs = sr^{-1}$
therefore, $r^2 = 1$ is equivalent to $rs = sr^{-1}$
therefore $\{x_i, y_i \mid x_i^2 = y_i^2 = 1, (x_i y_i)^2 = 1\}$ is exactly D_4 .